

# SOFTWARE REGISTRATION DOCUMENTATION

## KECOSO SMART DAIRY PLATFORM

|                       |                                      |
|-----------------------|--------------------------------------|
| Software Name:        | KECOSO Smart Dairy Platform          |
| Version:              | 1.0.0                                |
| Developer/Company:    | KECOSO Smart Dairy Team              |
| Registration Date:    | October 22, 2025                     |
| Software Type:        | Web-based AI Dairy Management System |
| Programming Language: | Python (Django Framework)            |
| Database:             | SQLite/PostgreSQL                    |
| License Type:         | Proprietary Commercial License       |

## ABSTRACT

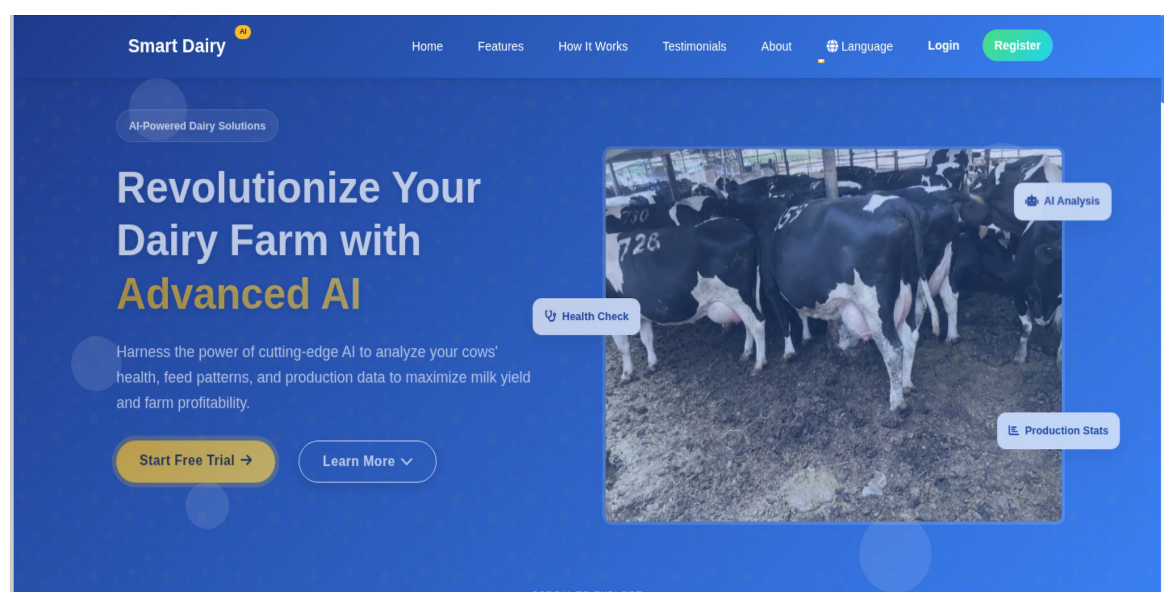
KECOSO Smart Dairy Platform is an innovative AI-powered web application designed to revolutionize dairy farm management in Kenya and East Africa. The system integrates advanced artificial intelligence, computer vision, and data analytics to provide comprehensive solutions for:

- AI-powered disease detection through image analysis
- Personalized feed optimization based on individual cow characteristics
- Bilingual AI assistant supporting English and Swahili languages
- Real-time health monitoring and production analytics
- Predictive modeling for milk yield optimization

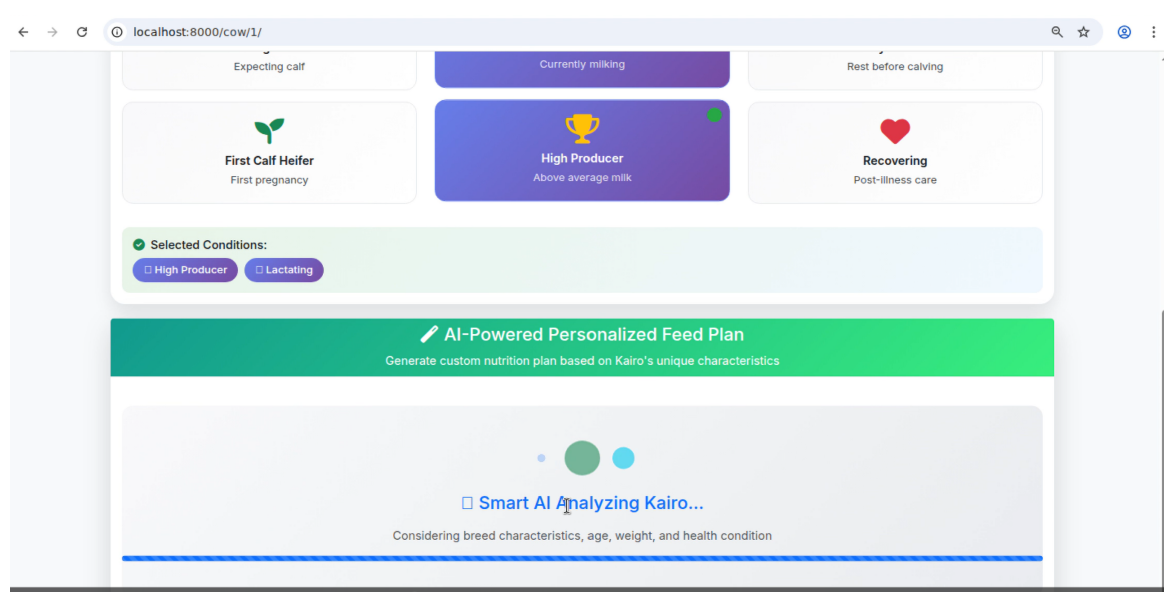
The platform addresses critical challenges in dairy farming by providing farmers with intelligent, data-driven insights to improve cow health, optimize feed costs, and maximize milk production efficiency.

# SYSTEM ARCHITECTURE & FEATURES

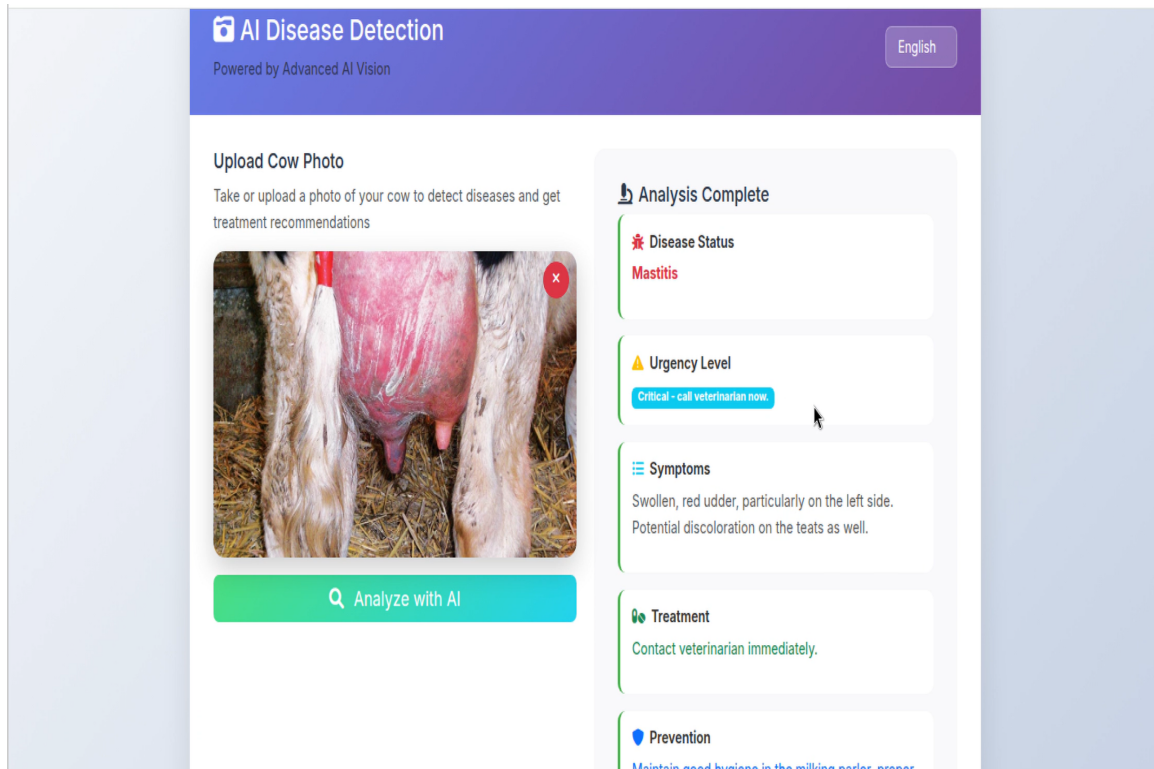
## 1. Main Dashboard Interface



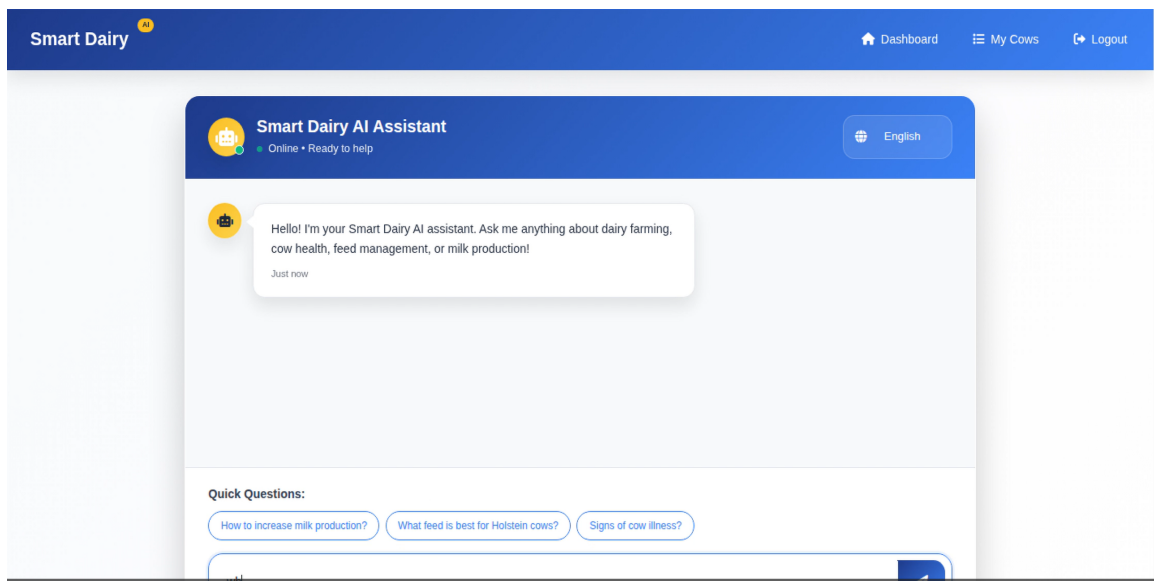
## 2. AI-Powered Feed Personalization



### 3. AI Disease Detection System



### 4. Bilingual AI Assistant





# TECHNICAL IMPLEMENTATION

## ***Core Application Structure (Django Views):***

```
from django.shortcuts import render, redirect from django.contrib.auth.decorators
import login_required from .models import Cow, HealthRecord, ProductionRecord from
.gemini_service import AIService @login_required def dashboard(request): user_cows =
Cow.objects.filter(farmer=request.user) total_cows = user_cows.count() avg_production =
user_cows.aggregate(Avg('daily_milk_yield')) context = { 'total_cows': total_cows,
'avg_production': avg_production['daily_milk_yield__avg'] or 0, 'cows': user_cows[:5] #
Recent 5 cows } return render(request, 'main/dashboard.html', context) def
ai_disease_detection(request): if request.method == 'POST' and
request.FILES.get('cow_image'): image = request.FILES['cow_image'] ai_service =
AIService() analysis = ai_service.analyze_cow_health(image) return
JsonResponse(analysis) return render(request, 'main/disease_detection.html')
```

## AI SERVICE INTEGRATION

```
import google.generativeai as genai from PIL import Image as PILImage class AIService:
def __init__(self): genai.configure(api_key=settings.GEMINI_API_KEY) self.model =
genai.GenerativeModel('gemini-pro-vision') def analyze_cow_health(self, image_file):
image = PILImage.open(image_file) prompt = '''Analyze this cow image for health
conditions. Identify any visible diseases, symptoms, and provide treatment
recommendations.''' response = self.model.generate_content([prompt, image]) return {
'disease_detected': self.extract_disease(response.text), 'symptoms':
self.extract_symptoms(response.text), 'treatment':
self.extract_treatment(response.text), 'urgency_level':
self.assess_urgency(response.text) } def generate_feed_plan(self, cow_data): prompt =
f'''Create personalized feed plan for cow: Breed: {cow_data['breed']} Weight:
{cow_data['weight']}kg Age: {cow_data['age']} months Production:
{cow_data['milk_yield']}L/day''' response = self.model.generate_content(prompt) return
self.parse_feed_recommendations(response.text)
```

# **COPYRIGHT & INTELLECTUAL PROPERTY**

## **Copyright Notice:**

© 2025 KECOSO Smart Dairy Team. All rights reserved.

## **Intellectual Property Rights:**

This software and all associated documentation, algorithms, user interfaces, and methodologies are the exclusive intellectual property of KECOSO Smart Dairy Team.

## **Patent Pending:**

AI-powered dairy management algorithms and computer vision disease detection methods.

## **Trademark:**

"KECOSO Smart Dairy" and associated logos are trademarks of KECOSO Smart Dairy Team.

## **Contact Information:**

Email: [info@kecososmartdairy.com](mailto:info@kecososmartdairy.com)

Phone: +254-XXX-XXXXXX

Address: Nairobi, Kenya